



Business Requirements Document (BRD)

Automated AI Travel Plan Creator

Version: 1.0

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Date: July 2026

1. Executive Summary

The objective of this project is to develop an AI-powered Travel Plan Creator capable of generating personalized multi-day travel itineraries based on user preferences.

Instead of manually searching flights, hotels, restaurants, tourist attractions, and calculating travel expenses, users should be able to provide a few travel preferences and receive a complete vacation plan within minutes.

The system will leverage AI models along with real-time search tools to recommend travel options, optimize itineraries, and estimate total trip costs while adhering to the user's preferences and budget.

2. Business Problem

Planning a vacation is often time-consuming and fragmented across multiple platforms.

Users typically need to:

- Search flights
- Compare hotels
- Find tourist attractions
- Research restaurants
- Calculate travel expenses
- Organize daily itineraries

This process requires significant manual effort and often results in inconsistent planning.

There is currently no single intelligent assistant capable of performing all these tasks together while considering personalized constraints.

3. Business Objectives

The proposed solution aims to:

- Reduce travel planning time.
- Provide personalized travel experiences.
- Generate complete itineraries automatically.
- Optimize travel plans according to budget.
- Simplify decision-making for travelers.



4. Business Goals

The system should:

- Generate customized travel itineraries.
- Recommend suitable flights.
- Recommend accommodations.
- Suggest sightseeing locations.
- Recommend restaurants based on dietary preferences.
- Calculate estimated travel expenses.
- Re-optimize the itinerary if the budget is exceeded.
- Allow users to modify generated plans.

5. Scope

In Scope:

- User travel preference collection
- Flight recommendations
- Hotel recommendations
- Restaurant recommendations
- Tourist attraction recommendations
- Multi-day itinerary generation
- Budget estimation
- Budget optimization
- Itinerary modifications
- Chat-based interaction

Out of Scope: For phase 1

- Flight booking
- Hotel booking
- Payment gateway
- Visa processing
- Travel insurance
- User authentication
- Mobile application
- Offline functionality

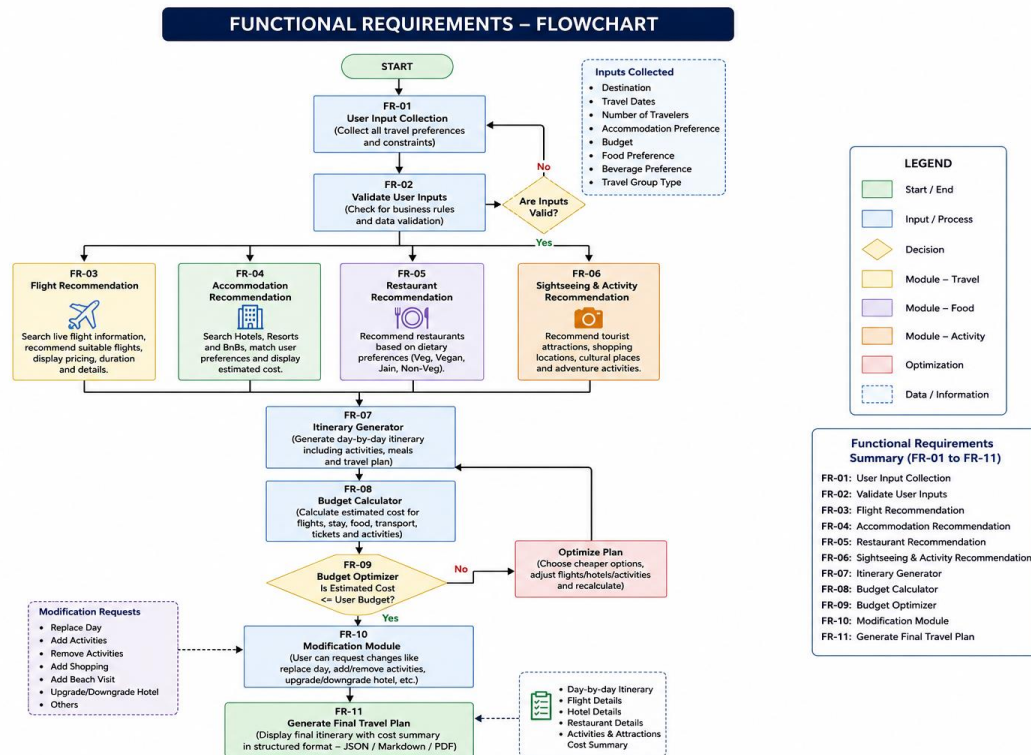
5. Stakeholders

Client, Business Analyst, AI Engineer, Backend Developer, QA Engineer, End Users.

6. Functional Requirements

- Collect destination, dates, passengers, accommodation, budget, dietary preferences, and travel group.
- Recommend flights.
- Recommend hotels/resorts/BNBs.
- Recommend restaurants matching dietary needs.
- Recommend sightseeing and shopping.
- Generate day-wise itinerary.

- Estimate total travel cost.
- Optimize recommendations if budget exceeds the limit.
- Support itinerary modifications.



7. Non-Functional Requirements

Fast response, scalable architecture, secure API handling, high availability, and user-friendly chat interface.

8. Business Rules

Validate dates and budget, respect dietary preferences, match travel dates with accommodation, avoid unsuitable recommendations for families, and remain within budget whenever possible.

9. Risks

Third-party API failures, inaccurate estimates, AI hallucinations, slow external services, and API costs.

10. Acceptance Criteria

System successfully generates personalized itineraries, recommendations, budget estimates, optimized plans, and supports itinerary modifications.